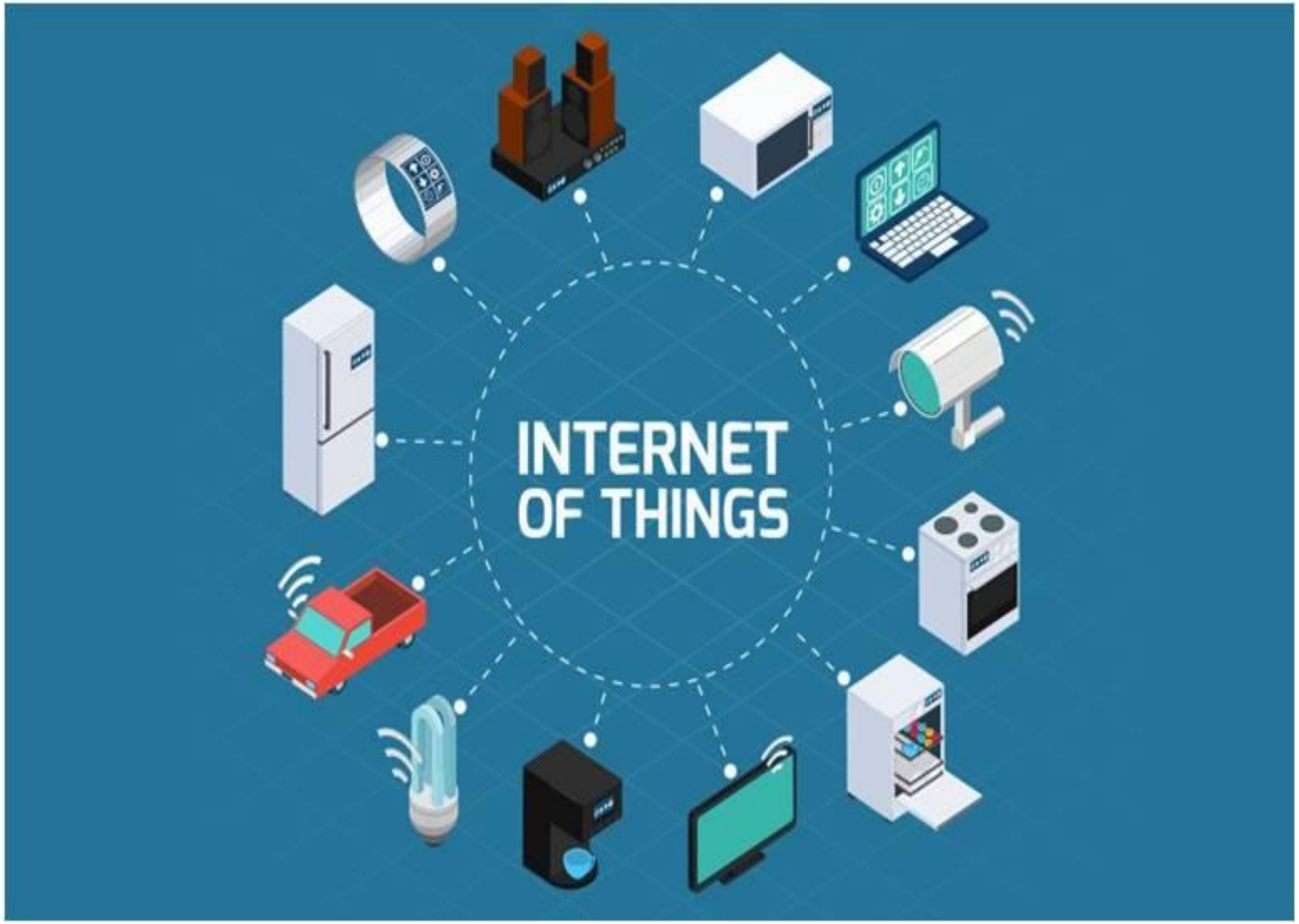
Introduction to Internet of Things

Lecture: 1 and 2

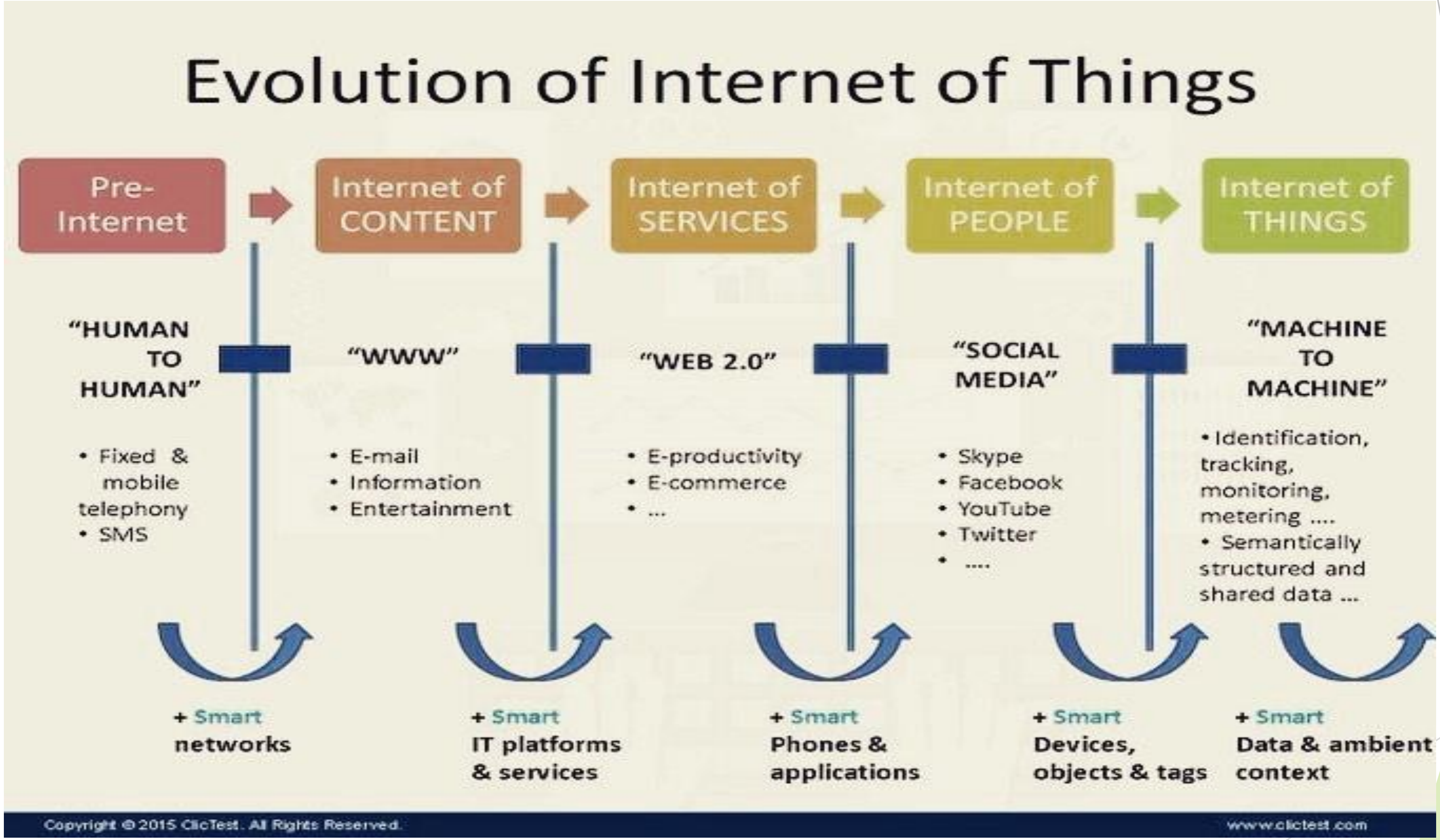
Internet of Things (IoT)

- Connecting various physical devices and objects throughout the world using internet.
- It allows objects to be sensed and controlled remotely across a network.
- Interconnection of everyday objects and devices through the internet, to collect, share and respond to data, making our lives more convenient, efficient, and smarter.


INTERNET
OF THINGS

An IoT System

- A system in which the computing devices such as sensors, actuators, embedded systems, and microcontrollers are interconnected via the internet or any other communication networks to perform intelligent tasks without human intervention.



Evolution of Internet of Things
Pre-Internet
Internet of CONTENT
Internet of SERVICES
Internet of PEOPLE
Internet of THINGS
"HUMAN TO HUMAN"
• Fixed & mobile telephony
• SMS
+ Smart networks
"WWW"
• E-mail
• Information
• Entertainment
+ Smart IT platforms & services
"WEB 2.0"
• E-productivity
• E-commerce
• ...
+ Smart Phones & applications
"SOCIAL MEDIA"
• Skype
• Facebook
• YouTube
• Twitter
• ....
+ Smart Devices, objects & tags
"MACHINE TO MACHINE"
• Identification, tracking, monitoring, metering ....
• Semantically structured and shared data ...
+ Smart Data & ambient context
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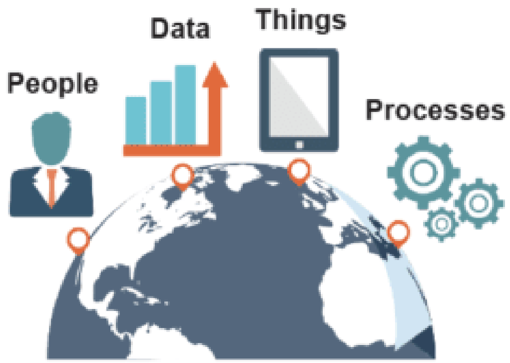
Machine-to-Machine

A device that captures an event and transmit the event over the network to an application. The application translates event to meaningful information.

Internet of Things

A network of uniquely identifiable things that communicate without human interaction using IP connectivity.

Internet of Everything



Bringing together the people, process, data, and things to make networked connections more meaningful by turning information into actions.

Intelligence in IoT

- IoT Devices are intelligent because they can sense, decide and act automatically.

Intelligence ≠ Artificial Intelligence

- An IoT device is intelligent because it can:

1. Sense the environment
2. Process data locally
3. Take action without human intervention
4. Communicate results

This is **automation**

Artificial Intelligence (AI)

AI-Based Intelligence (Advanced IoT)

- Learns from data
- Improves over time
- Uses ML models
- Example:

1. Learns when people usually come home
2. Turns lights ON before arrival

So, Simple IoT system has no Learning and Adaptation.

Automatic + Connected = Smart IoT

Learning + Prediction = AI based IoT

- An IoT system does not improve over time.
- They react the same way every time.
- No memory of past data.

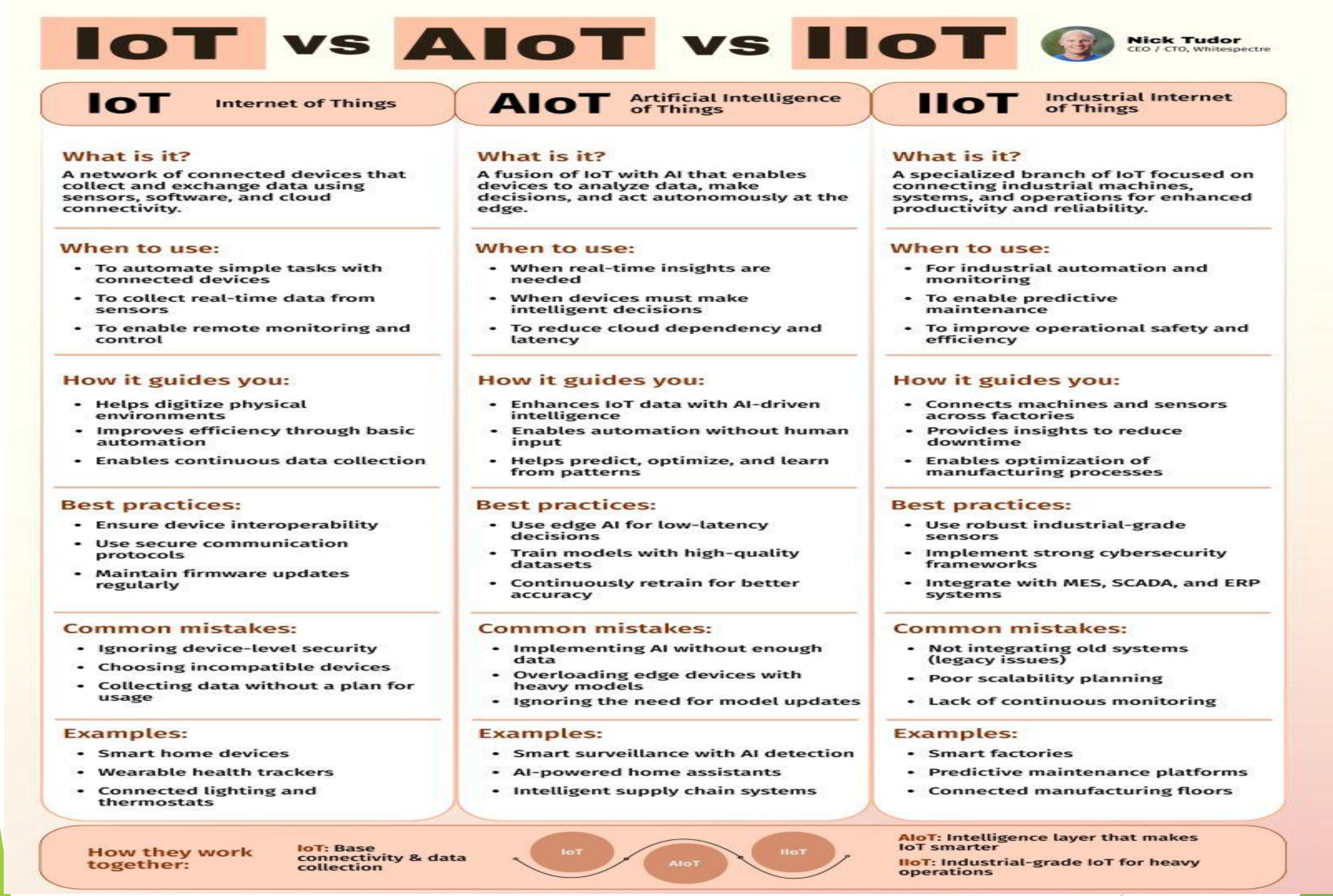
Simple IoT: *Sense → Decide (rules) → Act*

AI-IoT: *Sense → Learn → Predict → Act*

AI-based IoT System

- IoT system + AI
- Learning + adaptation
- Automation + connectivity
- Example:

  Wearable health monitoring with ML



IoT vs AIoT vs IIoT

Nick Tudor
CEO / CTO, Whitespectre

	IoT – Internet of Things	AIoT – Artificial Intelligence of Things	IIoT – Industrial Internet of Things
What is it?	A network of connected devices that collect and exchange data using sensors, software, and cloud connectivity.	A fusion of IoT with AI that enables devices to analyze data, make decisions, and act autonomously at the edge.	A specialized branch of IoT focused on connecting industrial machines, systems, and operations for enhanced productivity and reliability.
When to use:	• To automate simple tasks with connected devices • To collect real-time data from sensors • To enable remote monitoring and control	• When real-time insights are needed • When devices must make intelligent decisions • To reduce cloud dependency and latency	• For industrial automation and monitoring • To enable predictive maintenance • To improve operational safety and efficiency
How it guides you:	• Helps digitize physical environments • Improves efficiency through basic automation • Enables continuous data collection	• Enhances IoT data with AI-driven intelligence • Enables automation without human input • Helps predict, optimize, and learn from patterns	• Connects machines and sensors across factories • Provides insights to reduce downtime • Enables optimization of manufacturing processes
Best practices:	• Ensure device interoperability • Use secure communication protocols • Maintain firmware updates regularly	• Use edge AI for low-latency decisions • Train models with high-quality datasets • Continuously retrain for better accuracy	• Use robust industrial-grade sensors • Implement strong cybersecurity frameworks • Integrate with MES, SCADA, and ERP systems
Common mistakes:	• Ignoring device-level security • Choosing incompatible devices • Collecting data without a plan for usage	• Implementing AI without enough data • Overloading edge devices with heavy models • Ignoring the need for model updates	• Not integrating old systems (legacy issues) • Poor scalability planning • Lack of continuous monitoring
Examples:	• Smart home devices • Wearable health trackers • Connected lighting and thermostats	• Smart surveillance with AI detection • AI-powered home assistants • Intelligent supply chain systems	• Smart factories • Predictive maintenance platforms • Connected manufacturing floors

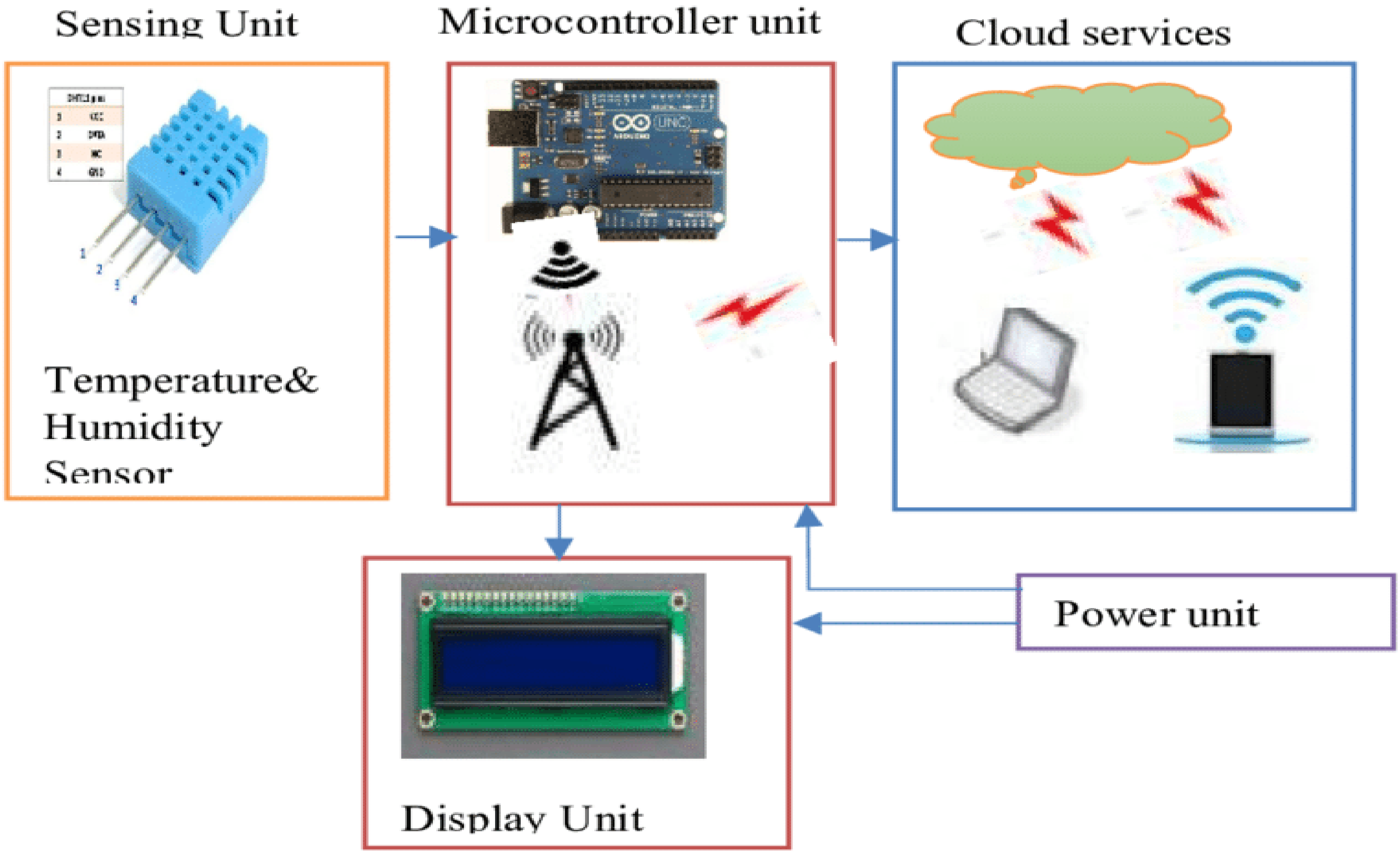
How they work together:

IoT: Base connectivity & data collection

IoT — AIoT — IIoT

AIoT: Intelligence layer that makes IoT smarter

IIoT: Industrial-grade IoT for heavy operations



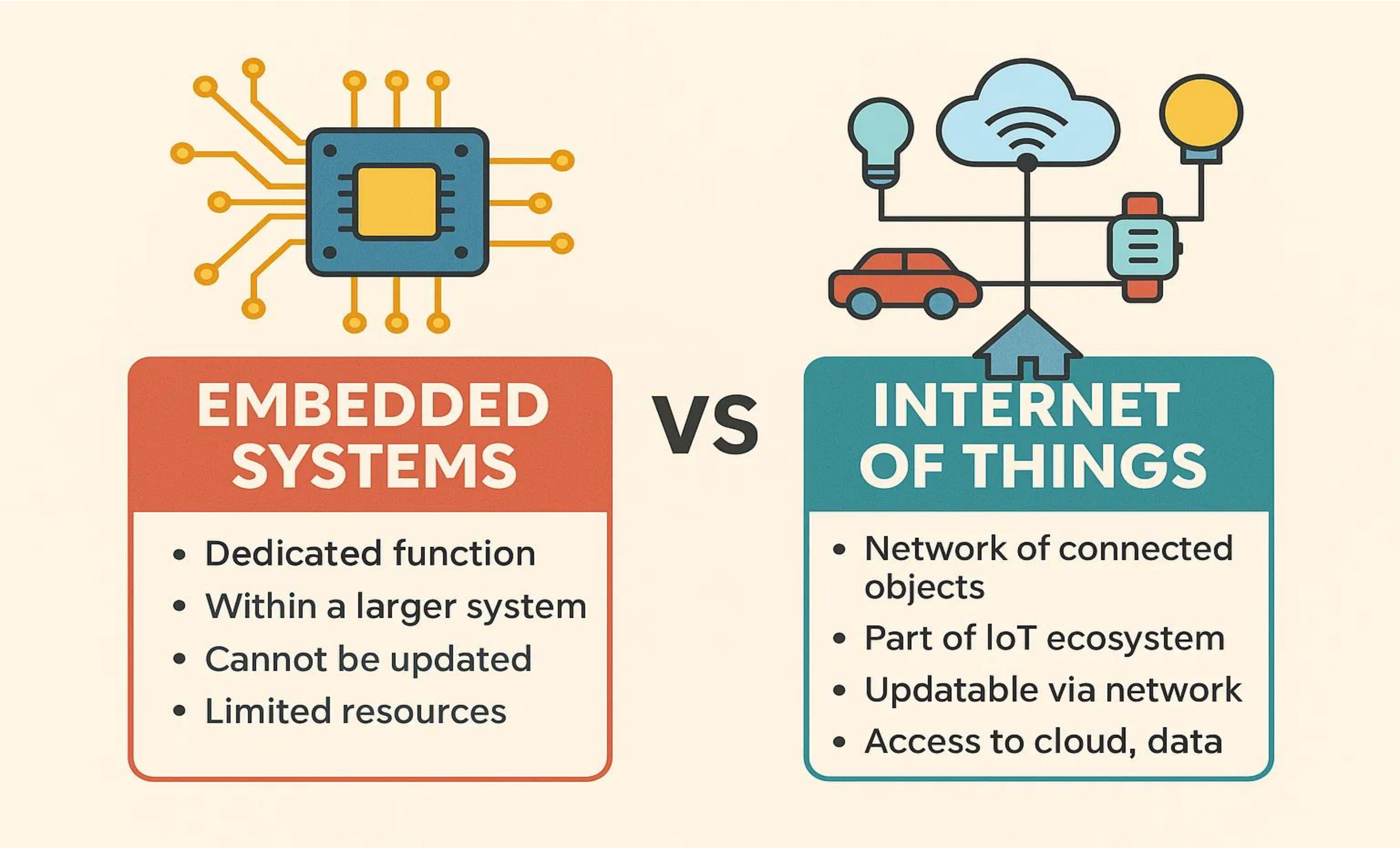
Sensing Unit
Microcontroller unit
Cloud services
Temperature&
Humidity
Sensor
Power unit
Display Unit

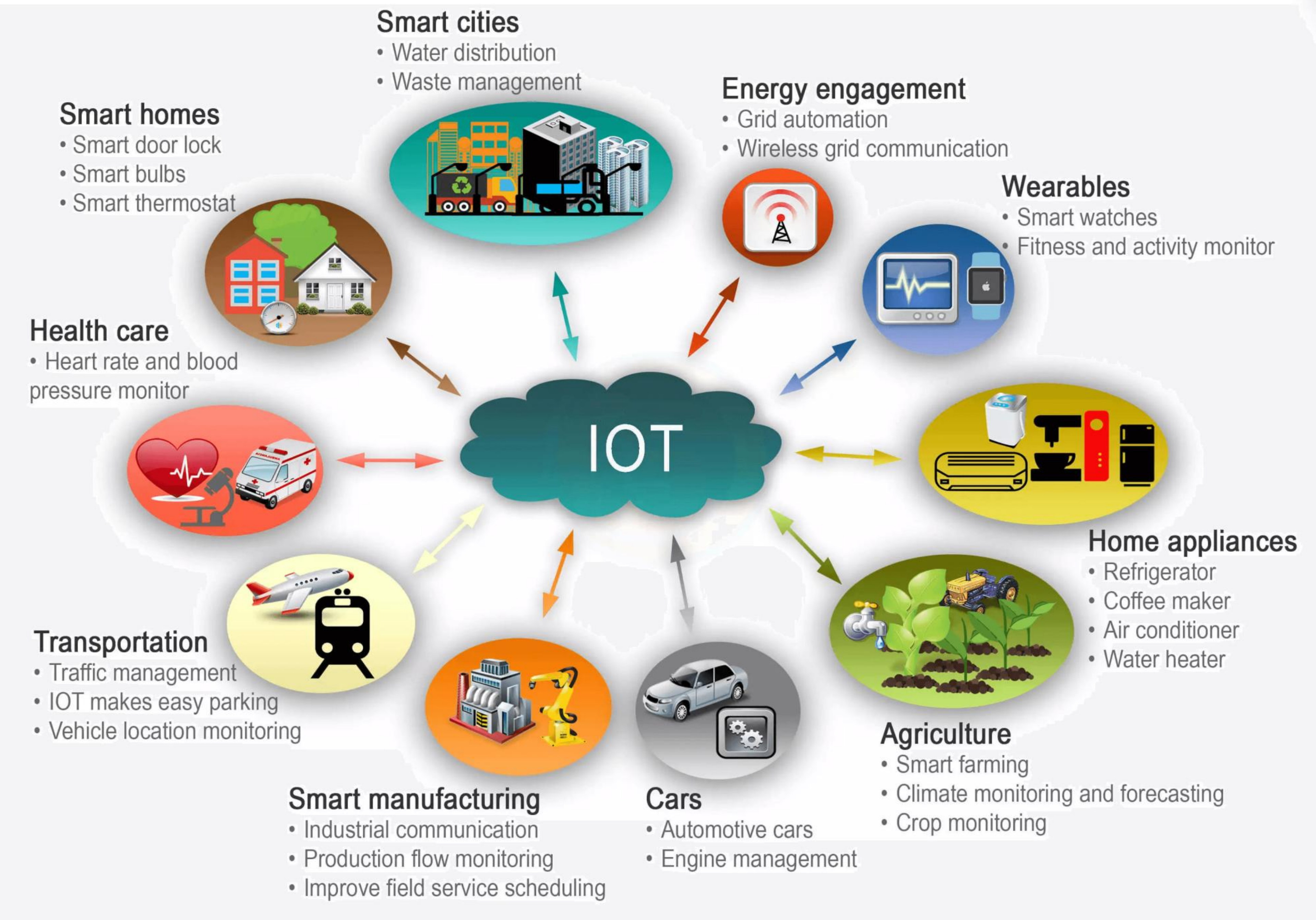
Related areas of IoT

- IoT is interdisciplinary

- Embedded Systems
- Computer Networks
- Wireless Communication
- Cloud Computing
- Data Analytics
- AI /ML (optional)

Every IoT device is an Embedded System (ES). **(ES + internet connectivity = IoT)**


EMBEDDED SYSTEMS
• Dedicated function
• Within a larger system
• Cannot be updated
• Limited resources
VS
INTERNET OF THINGS
• Network of connected objects
• Part of IoT ecosystem
• Updatable via network
• Access to cloud, data


Smart cities
• Water distribution
• Waste management
Smart homes
• Smart door lock
• Smart bulbs
• Smart thermostat
Energy engagement
• Grid automation
• Wireless grid communication
Wearables
• Smart watches
• Fitness and activity monitor
Health care
• Heart rate and blood pressure monitor
IOT
Home appliances
• Refrigerator
• Coffee maker
• Air conditioner
• Water heater
Transportation
• Traffic management
• IOT makes easy parking
• Vehicle location monitoring
Agriculture
• Smart farming
• Climate monitoring and forecasting
• Crop monitoring
Smart manufacturing
• Industrial communication
• Production flow monitoring
• Improve field service scheduling
Cars
• Automotive cars
• Engine management

Vision of IoT

- The vision of the Internet of Things (IoT) is to create a smart, connected world where physical objects are:

1. Identifiable
2. Connected
3. Intelligent
4. Autonomous

- The vision of IoT is to create a smart and connected environment where physical objects are uniquely identifiable, connected through communication networks, intelligent enough to analyze data, and autonomous in performing actions without human intervention.

IoT Vision

- Goals of IoT vision:

1. Seamless interaction between humans, machines, and the environment
2. Real-time data collection and decision-making
3. Automation to improve efficiency, safety, and quality of life
4. Integration of AI, cloud, and edge computing with physical systems

IoT aims to make the physical world digitally intelligent.

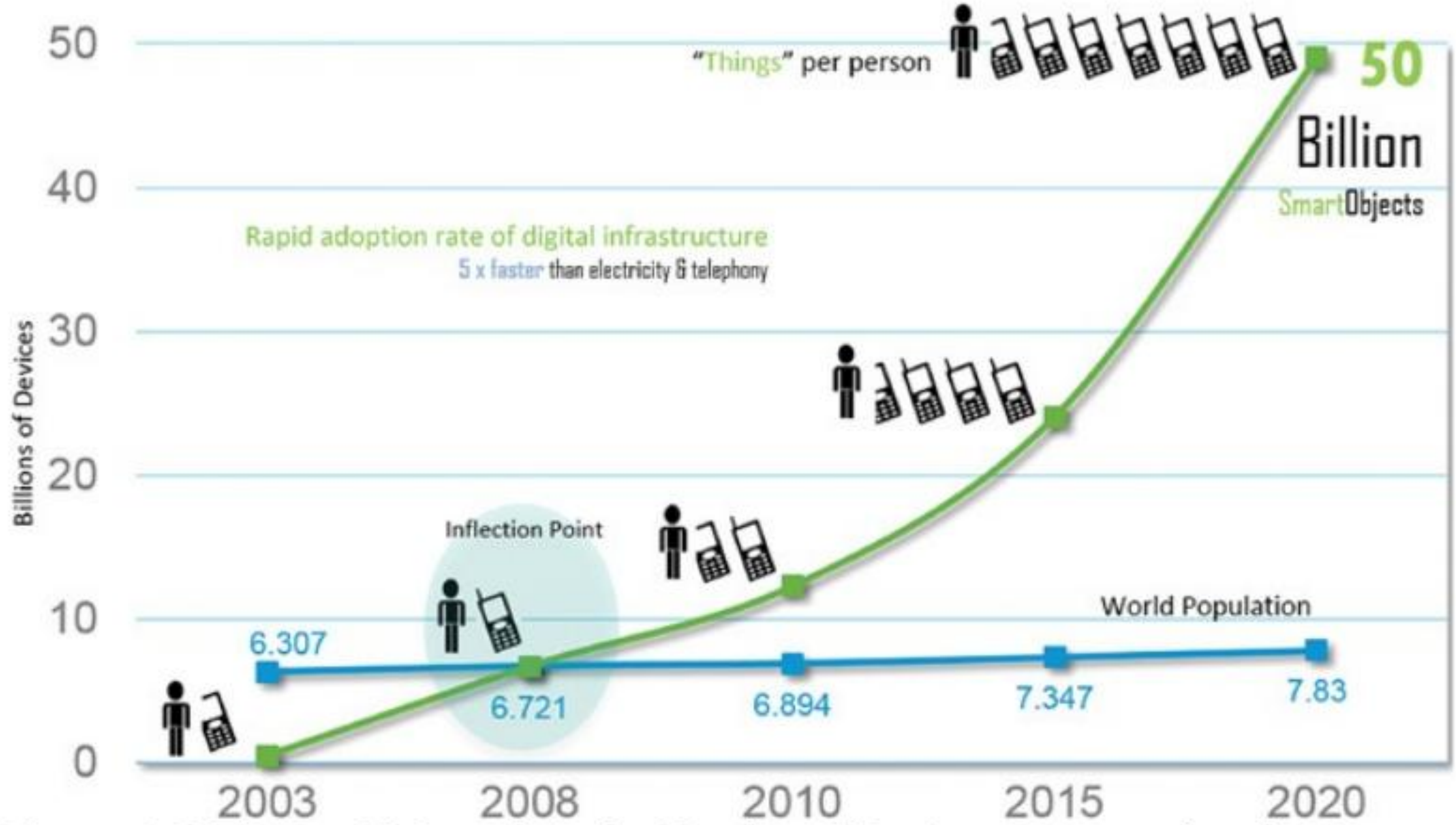


Figure 1-2 *The Rapid Growth in the Number of Devices Connected to the Internet*

Growth of IoT

- The growth of IoT has been rapid due to advancements in multiple technologies.

Technological growth:

a. Low-cost sensors and microcontrollers
b. Faster communication (4G, 5G, LPWAN)
c. Cloud computing and big data analytics
d. AI and machine learning integration
e. Edge computing for real-time processing

Growth of IoT

Application growth:

IoT is expanding across many domains:

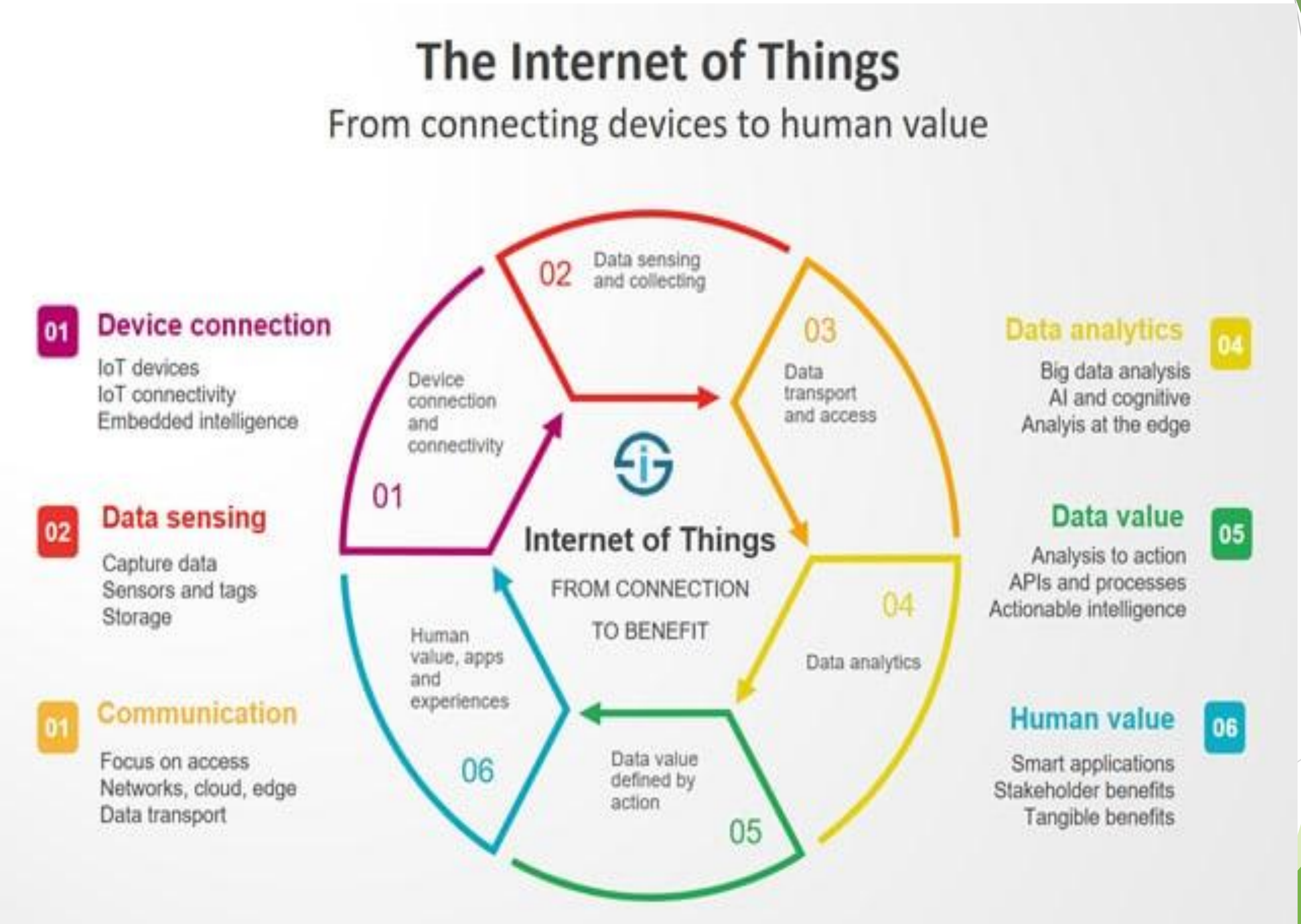
a. Smart homes – lighting, HVAC, security
b. Smart cities – traffic control, waste management
c. Healthcare – remote patient monitoring, wearables
d. Industry 4.0 – predictive maintenance, automation
e. Agriculture – smart irrigation, crop monitoring
f. Energy – smart grids and smart meters

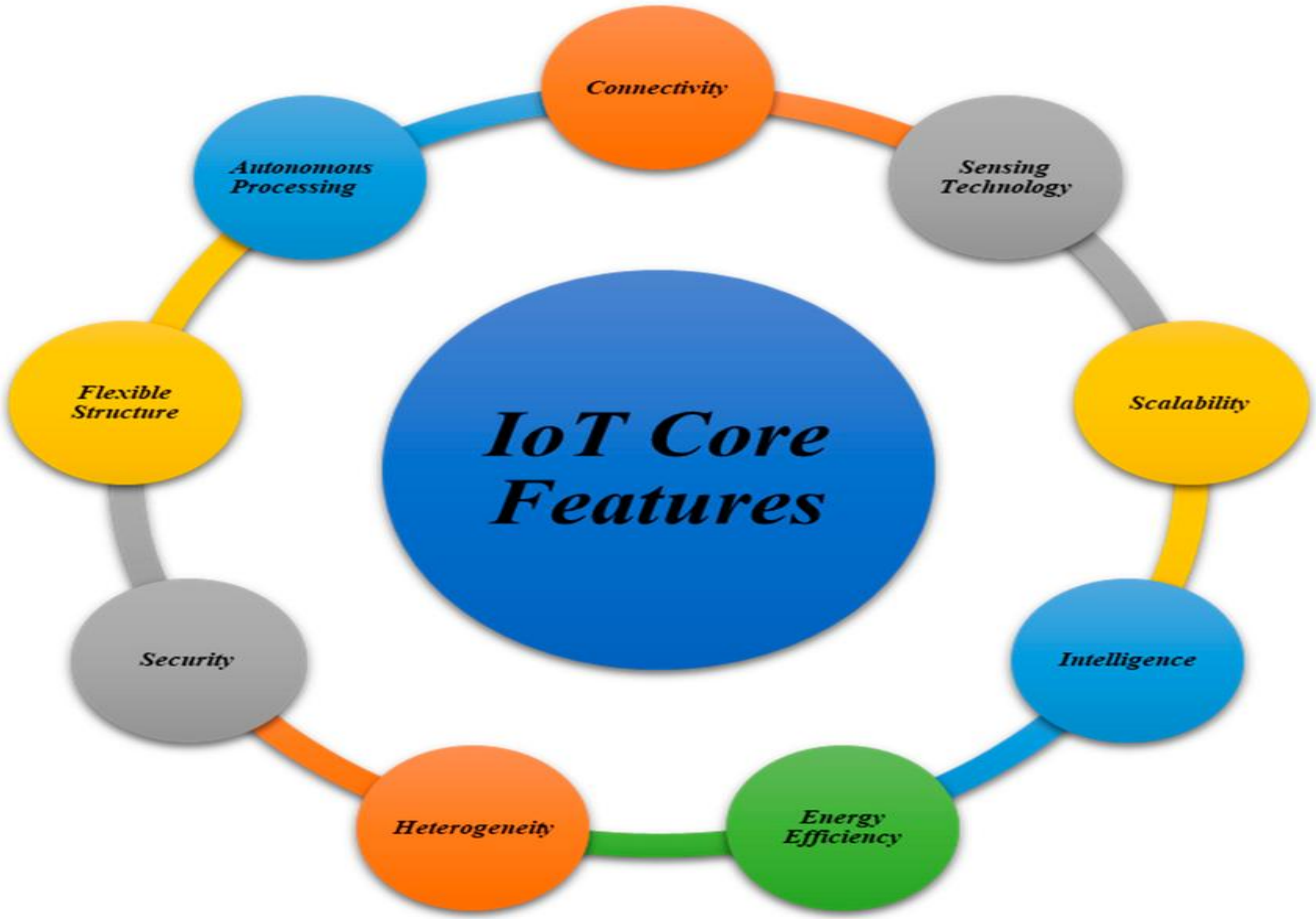
Market and Economic growth:

a. Massive increase in connected devices
b. Businesses shifting to data-driven models
c. Reduced operational costs
d. Improved productivity and decision-making

Social and Environmental growth:

a. Improved quality of life
b. Better healthcare access
c. Energy efficiency
d. Sustainable resource usage


The Internet of Things
From connecting devices to human value
01 Device connection
IoT devices
IoT connectivity
Embedded intelligence
02 Data sensing
Capture data
Sensors and tags
Storage
01 Communication
Focus on access
Networks, cloud, edge
Data transport
02 Data sensing and collecting
03 Data transport and access
01 Device connection and connectivity
Internet of Things
FROM CONNECTION TO BENEFIT
04 Data analytics
06 Human value, apps and experiences
05 Data value defined by action
Data analytics 04
Big data analysis
AI and cognitive
Analyis at the edge
Data value 05
Analysis to action
APIs and processes
Actionable intelligence
Human value 06
Smart applications
Stakeholder benefits
Tangible benefits


Connectivity
Sensing Technology
Scalability
Intelligence
Energy Efficiency
Heterogeneity
Security
Flexible Structure
Autonomous Processing
IoT Core Features

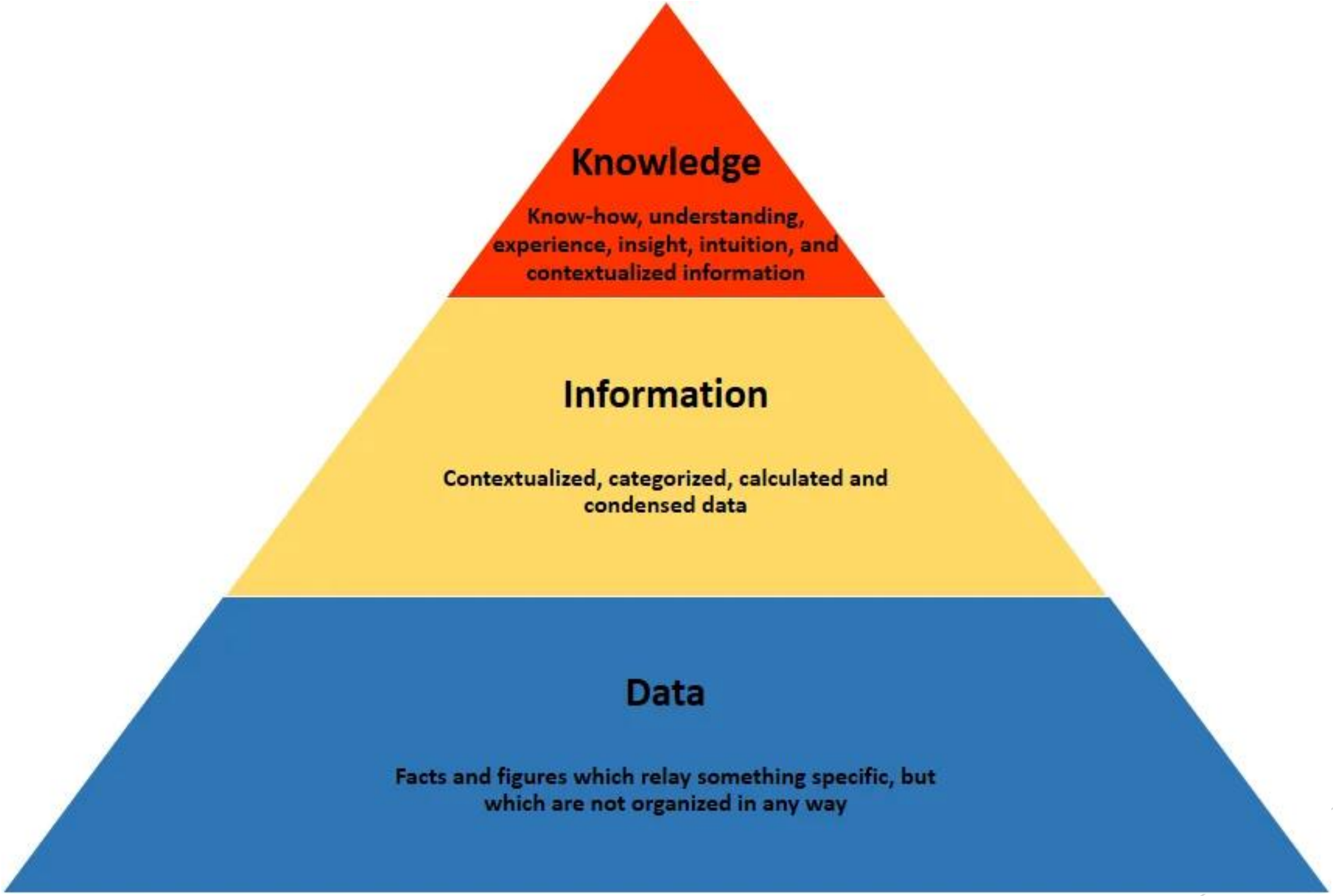
Issues and Challenges

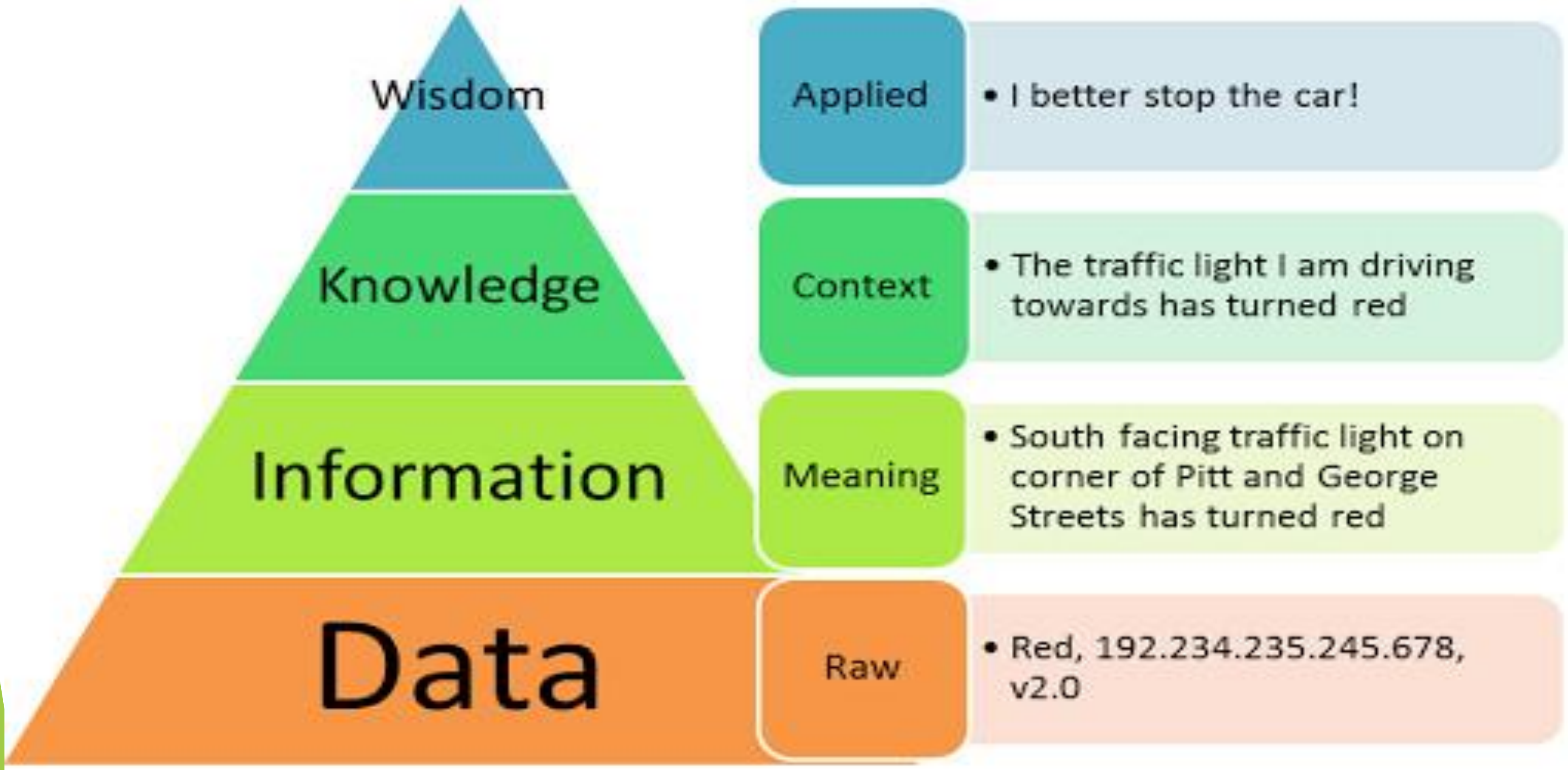
1. Security and Privacy
2. Interoperability and Standards
3. Power Consumption and Energy Management
4. Data Management and Storage
5. Network Connectivity and Latency
6. Scalability
7. Cost and Complexity

Advantages and Disadvantages of IoT

Advantages	Disadvantages
Minimizes the human work and effort	Increased privacy concerns
Saves time and effort	Increased unemployment rates
Good for personal safety and security	Highly dependent on the internet
Useful in traffic and other tracking or monitoring systems	Lack of mental and physical activity by humans leading to health issues.
Beneficial for the healthcare industry	Complex system for maintenance
Improved security in homes and offices	Lack of security
Reduced use of many electronic devices as one device does the job of a lot of other devices	Absence of international standards for better communication

Data -> Information -> Knowledge




Wisdom
Knowledge
Information
Data
Applied
• I better stop the car!
Context
• The traffic light I am driving towards has turned red
Meaning
• South facing traffic light on corner of Pitt and George Streets has turned red
Raw
• Red, 192.234.235.245.678, v2.0